We get $m^2 + 2m + 2 = 0$

The values of m (slopes) are not real (complex number), therefore no line will exist with the joint equation $2x^2 + 2xy + y^2 = 0$

We sometimes say that this equation represents imaginary lines.

Note that in the entire plane, only (0, 0) satisfies this equation

6.5.2 Angle between Pair of Straight Lines

Consider the equation of a pair of straight lines passing through the origin as:

$$ax^2 + 2hxy + by^2 = 0 \tag{6.30}$$

Let m_1 and m_2 be the slopes of these two lines.

By dividing (6.30) by x^2 and substituting $\frac{y}{x} = m$

$$\text{we get,} \quad bm^2 + 2hm + a = 0$$

This quadratic in m will have its roots as m_1 and m_2.

Thus, m_1+$m_2 = \frac{-2h}{b}$ and $m_1 m_2 = \frac{a}{b}$

If the angle between the two lines is θ Then

$$\begin{aligned}
\tan\theta &= \left|\frac{m_2 - m_1}{1 + m_2 m_1}\right| \\
&= \left|\frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_2 m_1}\right| \\
&= \left|\frac{\sqrt{\left(\frac{-2h}{b}\right)^2 - 4\frac{a}{b}}}{1 + \frac{a}{b}}\right| \\
\tan\theta &= \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|
\end{aligned}$$

As a consequence of this formula, we can conclude that

1. The lines are real and distinct, if m_1 and m_2 are real and distinct, that is if $h^2 - ab > 0$
2. The lines are real and coincident, if m_1 and m_2 are real and equal, that is if $h^2 - ab = 0$
3. The lines are not real (imaginary), if m_1 and m_2 are not real, that is if $h^2 - ab < 0$

Also, we see that the lines represented by (6.30), are parallel (since both pass through the origin, the lines are coincident lines) if $\tan\theta = 0$, that is $h^2 - ab = 0$, and perpendicular if $\cot\theta = 0$ that is $a + b = 0$

Pair of straight lines	Condition for parallel	Condition for perpendicular
$ax^2 + 2hxy + by^2 = 0$	$h^2 - ab = 0,$	$a + b = 0$